JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 95, NO. B12, PAGES 19,199–19,211, NOVEMBER 10, 1990

PALEOMAGNETIC EVIDENCE FOR PROTEROZOIC TECTONISM IN THE KAPUSKASING STRUCTURAL ZONE, ONTARIO

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Abstract. The Kapuskasing Structural Zone (KSZ) in the Superior Province of north-central Ontario is thought to be an oblique cross-section through Middle Archean crust. Samples from 30 sites in granulite gneisses and anorthosites of the KSZ have been analyzed using alternating field and thermal demagnetization methods. Three remanent magnetization components were found with mean directions of: D=23°, I=50° (k=52, α_{95}=11°, N=5 sites) for the A component; D=73°, I= -68° (k=50, α_{95}=5°, N=18) for the B component; and D=332°, I= -59° (k=53, α_{95}=17°, N=3) for the C component. V. Constanzo-Alvarez and D. J. Dunlop found similar A and B components in less metamorphosed rocks from the same units to the immediate west. They attributed the A component to uplift at 2.55 Ga with subsequent 15° to 30° WNW tilt and the B component to thermochemical overprinting at 1.1 Ga as the KSZ was reactivated during Keweenawan rifting and volcanism. We also attribute the A component to uplift during the 2.55 Ga Kenoran Orogeny with concomitant 10°±5° WNW downward tilting before intrusion of the 2.45 Ga Matachewan dikes, to give its present paleopole of 33°E, 67° (dp=6°, dm=9°, N=13) using data from both studies. We attribute acquisition of the B component to uplift at about 1.88 Ga, as suggested by recent thermal modeling and tectonic syntheses, to give a paleopole of 124°W, 30°N (dp=7°, dm=8°, N=20). The C component is found in only three sites and may be either an hybrid of a 1.1 Ga Keweenawan CRM with the B component or a fault-block tilted B component near the Ivanhoe Lake cataclastic zone.

Introduction

The Kapuskasing Structural Zone (KSZ) [Thurston et al., 1977] is a NNE trending belt of high grade metamorphic and plutonic rocks that cuts across the Archean Superior Province in Ontario (Figure 1). It transects and offsets the E-W trending Superior subprovinces for about 400 km from the Proterozoic margin near Lake Superior to where it disappears beneath Phanerozoic sedimentary rocks of the Hudson Bay Lowlands. Pronounced aeromagnetic and gravity anomalies demark the KSZ [Wilson and Brisbin, 1965]. Hypotheses for its origin include an upwarp in the Conrad discontinuity [Wilson and Brisbin, 1965], a horst [Bennett et al., 1967]; McGlynn, 1970], a suture zone [Wilson, 1968; Gibb and Wilcott, 1971], a failed arm [Burke and Dewey, 1973], a sinistral transcurrent fault [Watson, 1980], and an uplifted crustal basement on a low-angle reverse fault or a listric east-verging thrust fault known as the Ivanhoe Lake cataclastic zone (ILCZ) [Percival and Card, 1983, 1985]. Paleomagnetic study of the granulite gneisses and the Shawmere anorthosite in the KSZ was undertaken to provide evidence bearing on the various tectonic hypotheses. Recently Constanzo-Alvarez and Dunlop [1988] reported paleomagnetic data from the adjacent area to the west or, in general, from slightly less metamorphosed but equivalent units that are farther from the ILCZ (Figure 2). The ILCZ forms the boundary between Wawa and Abitibi Subprovinces and marks the eastern side of the KSZ. Our results are similar to and complement those of Costanzo-Alvarez and Dunlop [1988]. However, our interpretation of the B component is very different. We ascribe its acquisition to a 1.9 Ga event rather than to a 1.1 Ga event. Also we find a different, but similarly tenous, C component.

Geology of the KSZ

Within the Wawa Subprovince (Figure 1) there is a gradational transition over 120 km from the low-grade Michipicoten greenstone belt near Wawa, through the intermediate-grade Wawa domal gneiss terrain to the high-grade granulitic rocks of the KSZ (Figure 2). Mineral assemblage geobarometers indicate uplift of about 3.3 kbar (about 11 km) for lower greenschist rocks near Wawa, of about 5.4 kbar (about 18 km) on the western edge of the KSZ, and of up to 8.4 kbar (about 28 km) next to the ILCZ on the eastern edge of the KSZ [Ayres, 1969; Percival, 1983, 1986]. This pattern led Percival and Card [1983, 1985] to propose that the KSZ represents an oblique section of uplifted middle and lower continental crust by reverse or listric thrust faulting on the ILCZ with a regional dip of 8°±2° WNW. Cook [1985] reported preliminary seismic reflection results indicating that the ILCZ dips at 35°-40° WNW down to a depth of about 10 km before flattening. More recently, West et al., [1989], using more detailed seismic reflection data, obtained a 20° WNW dip at the ILCZ with progressive flattening to horizontal on the west side of the KSZ.

The Wawa and Abitibi Subprovinces are very similar. Both have volcanic-rich sequences representing several extrusive episodes dated between 2743 and 2696 Ma by the U/Pb zircon method and several granitic intrusive episodes dated between 2888 and 2615 Ma [Nunes and Jensen, 1980; Nunes and Pyke, 1980; Krogh et al., 1982; Percival and Sage, 1984; Turek et al., 1982, 1984; Sullivan et al., 1985; Frarey and Krogh, 1986]. Both subprovinces are cut by N-NNW trending diabase dikes of the Matachewan-Hearst swarm [Ernst and Halls, 1984] that give a U/Pb zircon age of 2454±2 Ma [Heaman, 1988]. This

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Paper number 90JB00972
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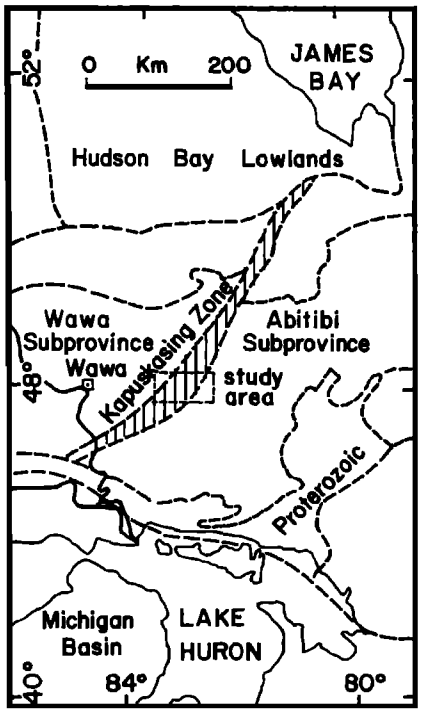


Fig. 1. Location of the KSZ study area in northern Ontario. OS and QS are the Opatika and Quetico subprovinces, respectively.

implies that the subprovinces are correlative with minimal transcurrent motion on the ILCZ and that they formed a stable craton by this time [Percival and Card, 1983]. These rocks were subsequently cut by the NE trending 2150±50 Ma Preissac tholeiitic dikes, the NW trending 1238±2 Ma Sudbury olivine diabase dikes and the ENE trending 1141±2 Abitibi olivine diabase dikes [Gates and Hurley, 1973; Van Schmus, 1975; Hanes and York, 1979; Krogh et al., 1988].

The KSZ itself is comprised mainly of tonalites, migmatitic paragneisses and mafic gneisses with a NNE foliation trend at low to intermediate WNW dips [Bennett et al., 1967; Thurston et al., 1977]. Garnet and orthopyroxene are the common indicators of granulite-grade metamorphism. The Shawmere anorthosite complex was emplaced into the gneisses and was metamorphosed to a high grade except in its central core where primary minerals and textures are sometimes preserved [Simmons et al., 1980; Riccio, 1981; Percival, 1983]. The minimum age for the gneisses and anorthosite is given by a U/Pb zircon date of 2.77 Ga on a garnetiferous tonalite, whereas younger ages from metamorphic zircons of 2.70 to 2.61 Ga from progressively closer to the ILCZ likely record cooling through 700°C upon uplift [Percival and Krogh, 1983; Constanzo-Alvarez and Dunlop, 1988; Percival et al., 1988]. Progressive cooling of the KSZ terrane is seen in other isotopic systems with the U/Pb sphene system closing between about 2.6 and 2.5 Ga, the K/Ar and $^{40}Ar/^{39}Ar$ hornblende systems between 2.55 and 2.4 Ga, the K/Ar and $^{40}Ar/^{39}Ar$ biotite systems between 2.35 and 2.2 Ga, and finally the Rb/Sr biotite system between 2.35 and 1.93 Ga, corresponding to about 300°C [Percival et al., 1988; Hunt and Roddick, 1987; Archibald et al., 1986]. The ENE trending Kapuskasing gabbro dikes were emplaced into this cooled KSZ terrane at 2043±14 Ma based on dating by conventional and laser $^{40}Ar/^{39}Ar$ methods [Hanes et al., 1988; Lee et al., 1988]. Two generations of alkalic syenite-carbonatite complexes of up to several kilometers in diameter

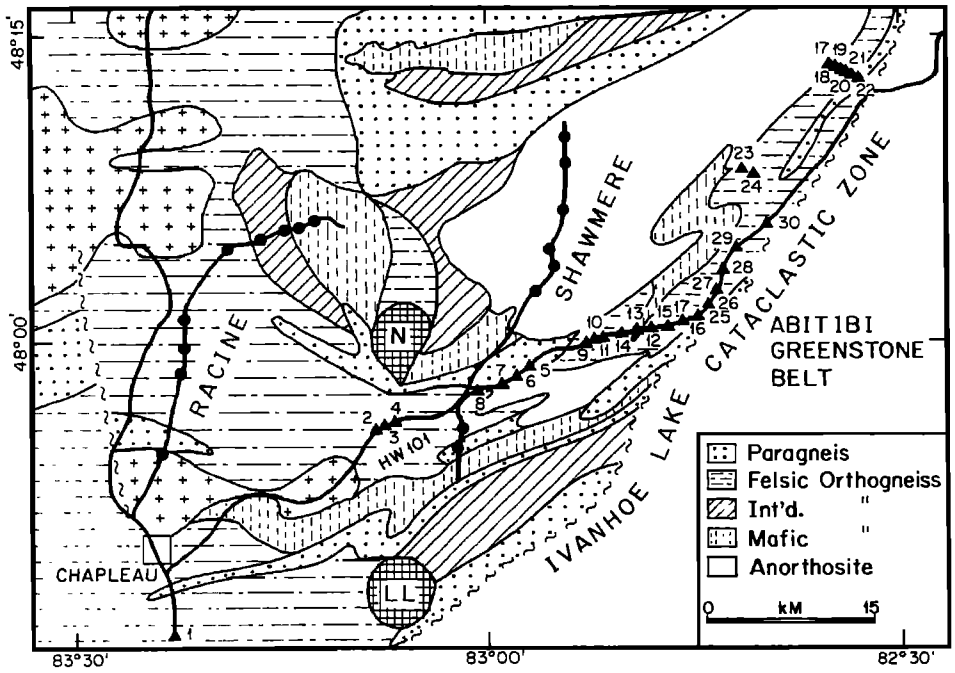


Fig. 2. Generalized geology of the study modified after Percival [1981] with the paleomagnetic site locations (triangles). The circles indicate the site locations on the Racine and Shawmere traverses of Constanzo-Alvarez and Dunlop [1988]. N and LL are the Nemegosenda and Lackner Lake carbonatite complexes.

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were emplaced into the KSZ. The older group includes the 1.87 Ga Borden Complex that is known from geophysics and drilling to be in the study area just northeast of Chapleau and the 1.89 Ga Cargill Complex to the north of the study area [Bell et al., 1987; L. Heaman, personal communication]. The younger group includes the 1.11 Ga Nemegosenda and 1.09 Ga Lackner Lake Complexes in the study area and the 1.08 Ga Shenango and 1.09 Ga Clay-Howells Complexes to the north [Gittins et al., 1967; Bell and Blenkinsop, 1980; L. Heaman, personal communication, 1988]. Thin lamprophyre dikes in the KSZ also appear to be part of this Keewanawan event and yield a K/Ar date of 1144±31 Ma [Stevens et al., 1982].

Experimental Methods

Five oriented cores were collected at each of 30 sites along a 60 km stretch of Highway 101 between Chapleau and Foleyet (Figure 2). This traverse is on the eastern side of the KSZ and dovetails with the traverses of Constanzo-Alvarez and Dunlop [1988]. Sampling was confined to the Shawmere anorthosite, to mafic to dioritic gneiss assemblages, and to more mafic bands in felsic gneiss assemblages of the host granulite gneisses. Three profiles of cores were collected for contact tests using a lamprophyre dike at site 16 and two Kapuskasing dikes at sites 26 and 28 and their country rocks. Two or more specimens were prepared from each core except for sites 3, 6, and 25. These sites yielded too few whole specimens for thermal demagnetization tests, consequently, no results are reported.

Magnetizations were measured using a Schonstedt SSM-1A spinner magnetometer, GSD-1 AF demagnetizer and TSD-1 thermal demagnetizer operated inside a magnetically shielded room. One specimen per site was demagnetized by alternating fields (AF) in 10 steps up to 100 mT and a second specimen was thermally demagnetized in 11 steps up to 640°C. Higher temperatures were employed in a few cases. The remaining specimens were demagnetized at 30 mT and again at 560°C because the pilot demagnetization tests showed that nearly all specimens had characteristic endpoints at these demagnetization levels. Using principal component analysis, a specimen's endpoint was accepted if its maximum angular deviation (MAD) was ≤10° [Kirschvink, 1980]. For those specimens demagnetized at 30 mT and again at 560°C, the origin was used the as third point. If the MAD was ≥10° for a specimen, then its demagnetization circle was determined if possible, and both endpoint and remagnetization circle data were combined to determine the site mean direction [Bailey and Halls, 1984]. A site mean was accepted only if it was based on data from four or more specimens and had a radius 5 for its cone of 95% confidence (α_{95} [Fisher, 1953]) of ≤20°.

Paleomagnetic Results

Natural Remanent Magnetization

The natural remanent magnetization directions (NRM) of all specimens form one significant directional population at the 95% confidence level [Kamb, 1959; Stupavsky and Symons, 1982], which is indistinguishable from the present Earth's magnetic field (PEMF) direction (Figure 3a). This suggests that the NRM is dominated by a low-coercivity, viscous remanence (VRM) overprint in the PEMF direction. After removal of the VRM overprints, two distinct components were isolated. Component A is directed NNE and down with an intermediate inclination and component B is directed ENE and steeply upward. Both A and B components are found in specimens from both granulite gneiss and anorthosite sites, and they appear to correspond to the A and B components identified by Constanzo-Alvarez and Dunlop [1988]. A third component, C, may be present that is directed NNW and upward at an intermediate inclination. It has an entirely different direction from the C component reported by Constanzo-Alvarez and Dunlop [1988].

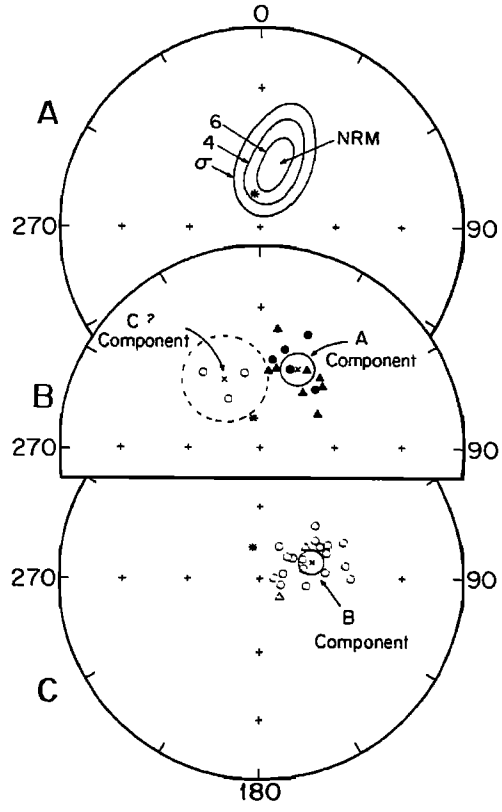


Fig. 3. Equal-area projections showing (a) percentage density contours for 235 NRM vector directions, where σ is the 95% confidence contour of 2.3% and the asterisk is the present Earth's magnetic field (PEMF) direction; (b) site mean directions for the A (solid symbols, down directions) and C (open symbols, up directions) components with circles from this study, triangles from Constanzo-Alvarez and Dunlop [1988] and crosses for the overall mean surrounded by its cone of 95% confidence [Fisher, 1953]; and (c) site mean directions for the B component with conventions as above.

A Component

The A component is common along the southwest end of the Highway 101 traverse near Chapleau, both away from the ILCZ and between the Racine and Shawmere traverses of Constanzo-Alvarez and

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Dunlop [1988]. After removal of VRM from the PEMF, the specimens show coherent univectorial decay of a single remanent magnetization component with high coercivities up to 100 mT and high laboratory unblocking temperatures up to the maximum unblocking value of 585°C for pure magnetite (Figure 4). Constanzo-Alvarez and Dunlop [1988] have discussed the magnetic characteristics of the A component in detail, having isolated it at a majority of their sites. We obtained similar results from the few sites in which we found the A component. One minor contradiction is that they argue in part that the A component is older than the B component because A is isolated only at high coercivities and unblocking temperatures typical of single domain-like magnetite whereas the multidomain fraction records only a VRM aligned with the PEMF. This generalization may be valid, however, in some of our specimens the A component is characterized by a distributed spectrum of unblocking temperatures above 350°C typical of some multidomain magnetite also (Figure 4b) to negate the argument. The A component site and unit mean statistics are given in Table 1 and shown in Figure 3b.

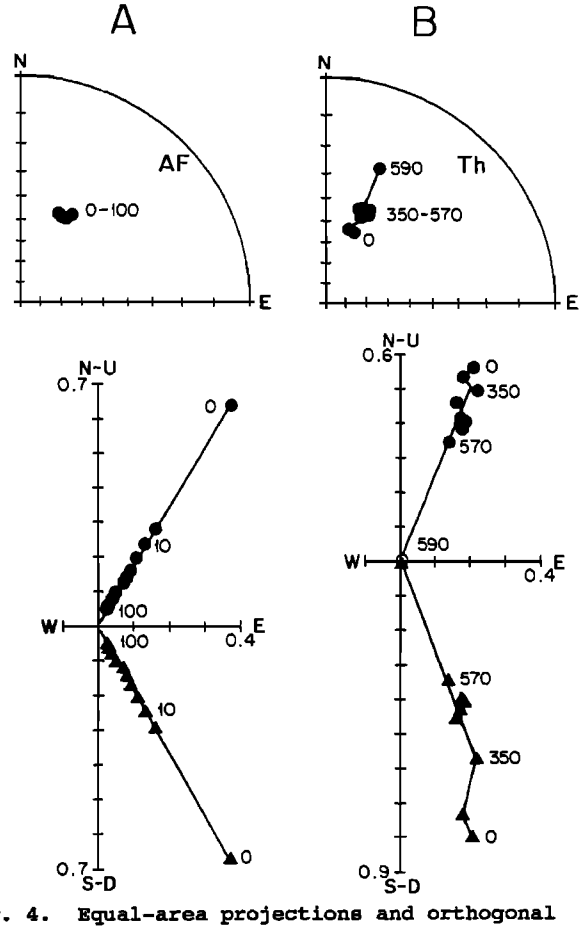


Fig. 4. Equal-area projections and orthogonal demagnetization plots of examples of the A component from (a) site 9 on AF demagnetization (AF) with a specimen NRM intensity (J_0) of 4.1 Am^{-1} and (b) site 8 (J_0=6.2 Am^{-1}) on thermal demagnetization (Th). Stereograms use solid (open) symbols for down (up) vectors. Orthogonal plots use circles (triangles) to depict the horizontal N-E-S-W (vertical Up-E-Down-W) plane. Axial lengths are expressed as a ratio of the NRM intensity. Demagnetization treatments are in mT (°C) on the AF (Th) plots.

B Component

The B component is common along the center and northeast end of the traverse, that is closer to the ILCZ. A majority of the anorthositic and mafic gneiss specimens contain a minor PEMF overprint that is first removed before a well-defined B component is isolated by both AF and thermal demagnetization (Figures 5c and 5d). A few of these specimens show the isolation of B as a univectorial component with coercivities up to 100 mT and unblocking temperatures up to 580°C (Figures 5a and 5b). The relatively few dioritic and intermediate to felsic gneissic specimens have a minor to substantial VRM overprint that aligns with either the A component or PEMF direction on initial demagnetization before the B direction is isolated (Figure 6). A small number of specimens show that the B direction is preserved in hematite, as evidenced by laboratory unblocking temperatures between 600 and 675° (Figure 7a). One specimen from site 17 exhibits a demagnetization trajectory that projects from the A or PEMF direction towards the B direction, but it could be isolating the C component of Constanzo-Alvarez and Dunlop [1988] (Figure 7b). The site and unit mean statistics are given in Table 2 (Figure 3c).

Table 1. Site Mean Directions for the A Component After Demagnetization

Site	N,E,R	D°	I°	k	α_{95}°	A
1	10,5,2	15	49	117	6	*
5	9,5,0	46	59	55	10	*
7	9,8,0	54	52	42	9	*
8	8,8,0	24	39	34	10	*
9	10,4,1	24	46	80	9	*δ
10	10,1,0	41	52			δ
21	20,2,2	14	48	13	27	δ
Mean	(5)	22.9	50.0	52	10.7	
CA:D	(6)Rac.	21.0	54.0	46	9.9	
CA:D	(2)Shaw.	50.0	60.2	104	24.8	
CA:D	(8)Comb.	27.3	56.2	39	9.0	
ALL	(13)	25.5	53.8	43	6.4	

N,E,R are the number of specimens demagnetized, giving used endpoints, and used remagnetization circles, respectively. Asterisks denote accepted (A) site means used to calculate the unit mean. Deltas indicate sites which also gave B component results (Table 2). CA:D data are accepted sites from Constanzo-Alvarez and Dunlop [1988] for the Racine (Rac.), Shawmere (Shaw.) and combined (Comb.) traverses, respectively. ALL is the overall unit mean direction from both studies.

C Component

Our C component is found in only three sites along the northeast end of the Highway 101 traverse near Foleyet. All three sites have a stable magnetization in magnetite that is mostly

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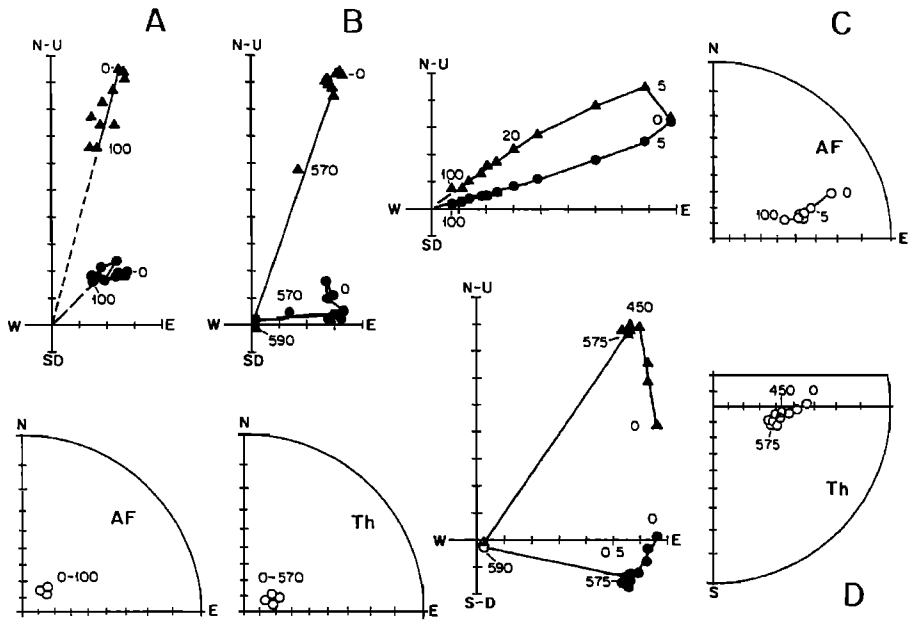


Fig. 5. Equal area projections and orthogonal demagnetization plots of examples of the B component from sites (a) 14 (J_0=5.0x10^{-2} Am^{-1}), (b) 14 J_0=4.9x10^{-2} Am^{-1}, (c) 23 (J_0=2.3 Am^{-1}), and (d) 16 (J_0=2.2x10^{-1}Am^{-1}). Conventions as in Figure 4. Note the initial northeasterly and down VRM component in Figures 5a and 5d.

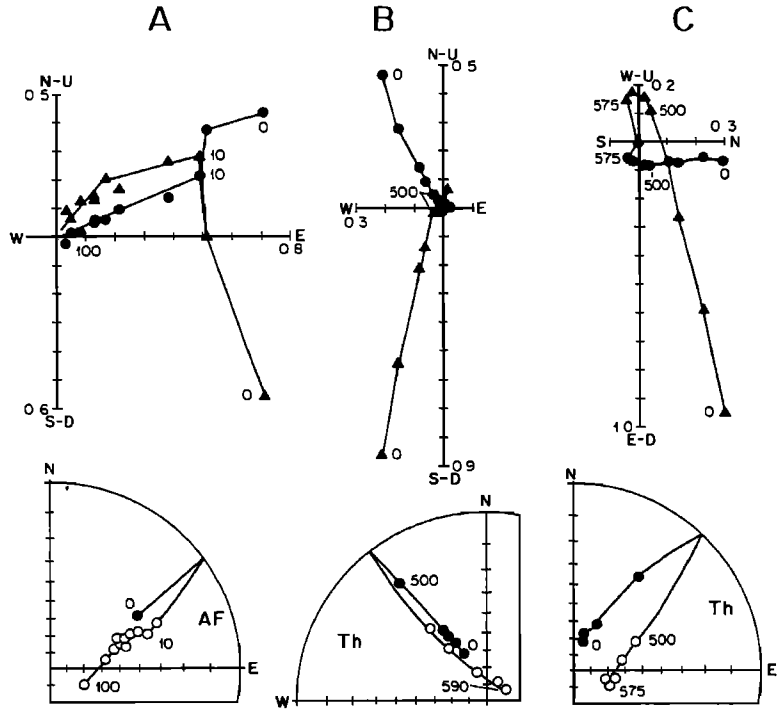


Fig. 6. Equal-area projections and orthogonal demagnetization plots of examples of bivectorial remanence with the A or PEMF components being first removed to isolate the B component from sites (a) 11 (J_0=9.5x10^{-1} Am^{-1}), (b) 21 (J_0=4.9x10^{-1} Am^{-1}) and (c) 28 (J_0=9.0 Am^{-1}). Conventions as in Figure 4.

unblocked between 545°C and 585°C following removal of a minor VRM by 350°C (Figure 8). It differs from the B component only in having a statistically significantly different direction (Table 3; Figure 3b).

Kapuskasing Dike Contact Tests

At sites 26 and 28, Kapuskasing gabbro dikes about 18 m wide strike ENE, dip very steeply southward and cut mafic granulite gneiss. At site 26, specimens from four samples in the dike,

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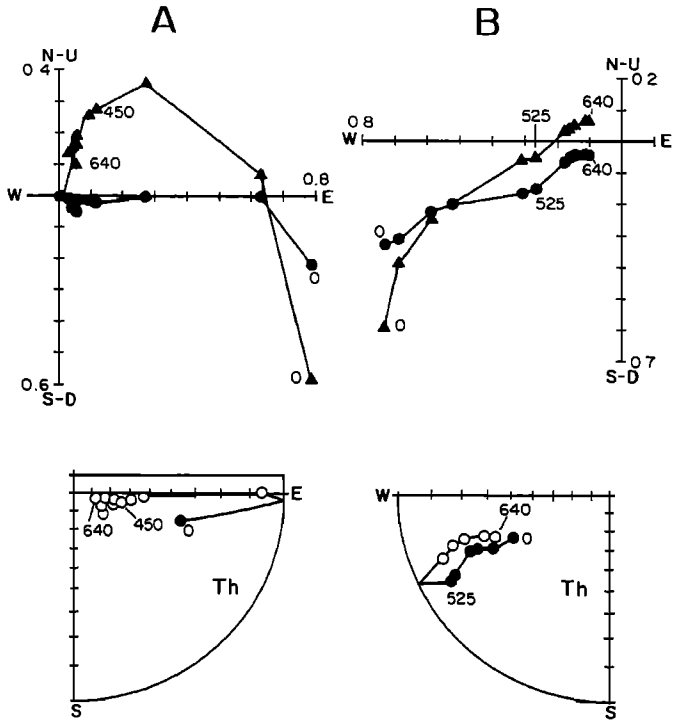


Fig. 7. Equal area projections and orthogonal demagnetization plots showing (a) the B magnetization being isolated in laboratory unblocking temperatures of over 585°C that are typical of hematite in a specimen from site 24 (J_0=2.1 Am^{-1}), and (b) the single possible specimen from site 17 (J_0=1.6 Am^{-1}) of the C component as described by Constanzo-Alvarez and Dunlop [1988]. Conventions as in Figure 4.

Table 2. Site Mean Directions for the B Component After Demagnetization

Site	N,E,R	D°	I°	k	α_{95}	A
9	10,2,2	82	-54	33	16	*
10	10,5,0	77	-71	41	12	*
11	10,1,6	67	-60	53	8	*
12	10,3,2	59	-74	32	14	*
13	10,4,3	90	-52	17	15	*
14	9,4,1	54	-75	46	11	*
15	10,8,1	99	-70	32	9	*
16	9,3,1	85	-63	30	17	*
17	9,2,6	33	-75	86	6	*
18	8,2,2	79	-68	50	13	*
20	10,4,4	65	-59	38	9	*
21	10,3,2	67	-71	56	10	*
22	10,5,5	57	-62	40	8	*
23	10,6,4	62	-62	24	10	*
24	10,8,1	66	-52	30	10	*
26	10,8,1	88	-80	30	10	*
28	10,6,4	102	-81	85	5	*
29	8,4,3	91	-84	52	8	*
Mean	(18)	72.7	-68.1	50	5.0	
CA:D	(2)	72.7	-76.9	26		
ALL	(20)	72.7	-69.0	47	4.8	

Conventions as in Table 1

from two samples within 2 m of the dike, and from two samples from 20 m 80 m away from the dike all exhibit the B component direction after demagnetization (Figure 9a). At site 28 a profile of 39 cores was collected perpendicular to the contact, however, an initial selection of eight cores gave the same negative test result as found at site 26 (Figure 9b). The results of these contact tests suggest that the B component was acquired at the time of intrusion of the Kapuskasing dikes or at some time thereafter as a metamorphic overprint.

Lamprophyre Dike Contact Test

At site 16, a 43 cm wide lamprophyre dike cuts tonalite host rocks, strikes ENE and dips very steeply. A 2 m long profile of cores was collected with seven cores in the dike and 25 cores in the host rock (Figure 10a). For most cores, one specimen was AF demagnetized and a second was thermally demagnetized. The dike specimens yield a WNW and moderately inclined down direction that is not recognized elsewhere in the collection (Figure 9b). The host rock is completely remagnetized next to the dike, exhibiting the dike direction up to 14 cm from the contact in unblocking temperatures up to 500°C. From 14 to about 50 cm the dike direction is found up to progressively lower unblocking temperatures above which the B component is found. Beyond 50 cm, only the PEMF and B directions are present. This positive contact test shows that the acquisition of the B component by the host rock predates intrusion of the lamprophyre dike. Following Schwarz [1977] [see also Symons et al., 1980; McClelland Brown, 1981], the data imply that the host rock had cooled to an ambient temperature of ≤200°C when the dike was intruded.

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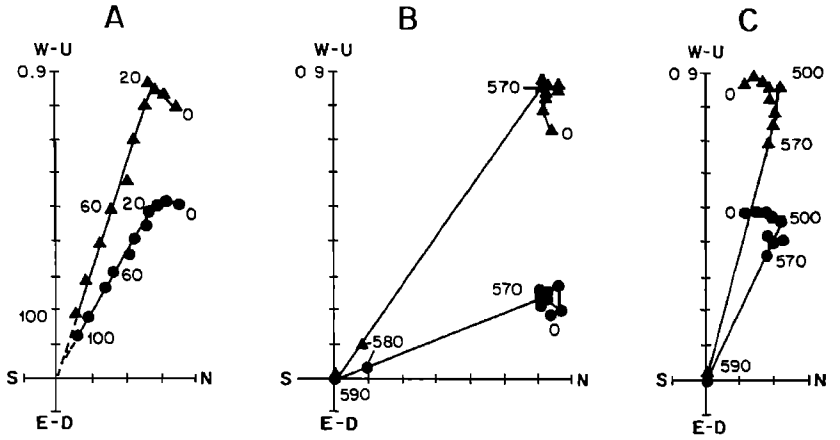


Fig. 8. Orthogonal demagnetization plots of examples of the C component from (a) site 27 (J_0=2.82 Am^{-1}) on AF demagnetization, (b) site 30 (J_0=1.98 Am^{-1}) on thermal demagnetization, and (c) site 19 (J_0=6.80x10^{-2} Am^{-1}) on thermal demagnetization. Conventions as in Figure 4.

Table 3. Site Mean Directions for the B Component After Demagnetization

Site	N,E,R	D°	I°	k	α_{95}	A
19	6,6,0	326	-66	26	13	*
27	8,7,0	323	-50	14	17	*
30	10,5,3	348	-58	141	5	*
Mean	(3)	332.1	-58.1	56	16.6	

Conventions as in Table 1.

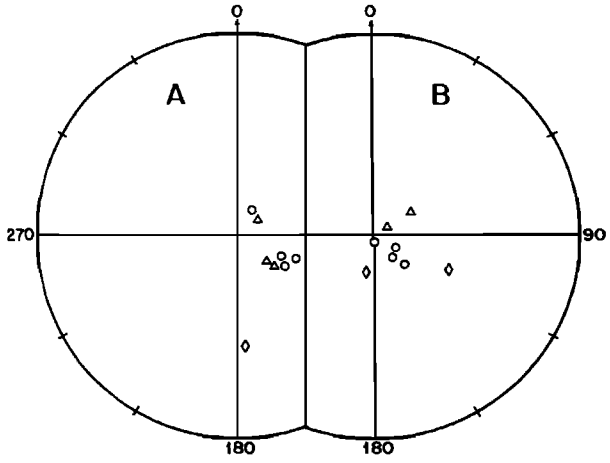


Fig. 9. Thermally and AF demagnetized remanence directions from Kapuskasing dike contact tests at sites (a) 26 and (b) 28. Dike directions are shown by circles, baked zone directions by diamonds, and remote host rock directions by triangles. Both contact tests are negative because all the directions form a single population.

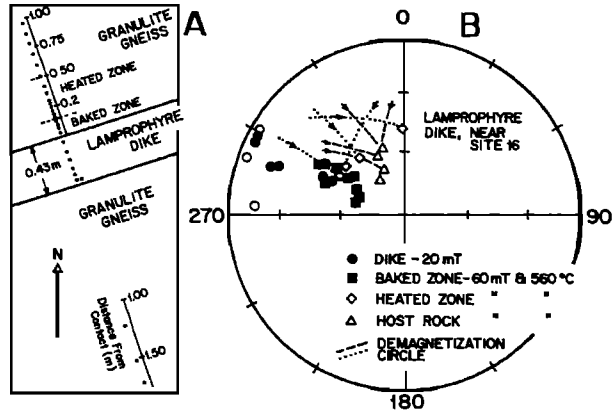


Fig. 10. Lamprophyre dike contact test showing (a) plan view of the sampling profile and (b) positive inclinations from the dike and baked zone after demagnetization compared with negative inclinations for the heated and remote host rock zones after removal of an initial down component to yield a positive contact test.

Discussion of the A Component

The A component is defined in five sites from the southeastern half of the Highway 101 traverse. Constanzo-Alvarez and Dunlop [1988] found it at eight sites on their Racine traverse and five sites on their Shawmere traverse. They believed that the two traverses gave different A component directions because of different amounts of tilt. However, when site means based on three or fewer specimens are excluded, as done in this study, the Racine and Shawmere traverses yielded only six and two A component site means, respectively, and these two populations are not significantly different at the 95% confidence level [McFadden and Lowes, 1981]. Also their overall A component mean is statistically the same as that found in this study (Table 1). Therefore all 13 accepted site means from both studies are grouped together as a single population with a mean direction of D=25.5°, I=53.8° (k=43, α_{95}=6.4°, N=13).

The A component is found in virtually every rock type in the area from massive Shawmere anorthosite to felsic granulite gneiss, and its direction is nearly perpendicular to the dominant

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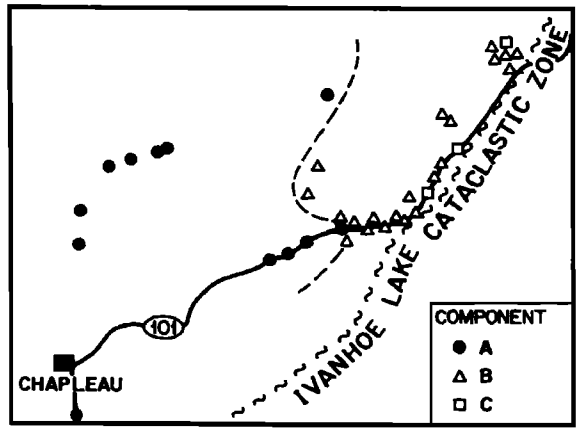


Fig. 11. Map of the KSZ study are as in Figure 1 showing the locations of sites with accepted A (solid circles), B (open triangles) and C (open squares) magnetization components from this study and from Constanzo-Alvarez and Dunlop [1988]. The dashed line separates A and B component sites.

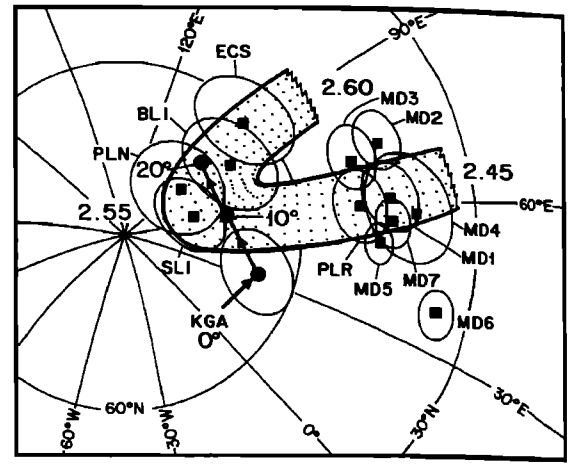


Fig. 12. APWP for the Superior Province from 2.70 and 2.45 Ga. The poles are CS, Chibougamau sills, 2.59 Ga (Rb/Sr) [Ueno and Irving, 1976]; BL1, Burchell Lake granite, about 2.58 Ga (indirectly dated) [Dunlop, 1984b]; SL1, Shelley Lake granite, 2.58±0.02 Ga ($^{40}Ar/^{39}Ar$) [Dunlop, 1984a]; PLR,PLN, Poobah Lake complex reverse and normal, 2.55 Ga (K/Ar isochron) [Dunlop, 1985] with both assumed to be related to the 2.55 Ga heating event; Matachewan diabase dikes, 2.45±0-.01 Ga (U/Pb zircon [Heaman, 1988]) with MD1 from Irving and Naldrett [1977], MD2 from Schutts and Dunlop [1981], MD3 from Ernst and Halls [1984], MD4 from Vandall and Symons [1987], and MD5, MD6, and MD7 for the CH-N, CH-R and RA groups in Halls and Palmer [1990]; and KGA is the Kapuskasing granulite A component pole of this study shown before (0°) and after 10° and 20° tilt corrections for KSZ thrusting. The pole, after the preferred 10° tilt correction is at 59°E, 71°N (dp=6°, dm=9°). The pole of the four Matachewan dikes for which Ernst and Halls [1984] suggested a 15° WSW tilt is covered by the MD7 pole.

NNE trending and WNW dipping gneissic foliation. This indicates that the remanence is not biased into the plane of the fabric.

Relative to the B component sites, the A sites are further to the west of the ILCZ (Figure 11). We suggest that the A component was progressively unblocked and replaced by the B component in response to the increasing metamorphic grade towards the ILCZ, and this implies that the A component predates the B component. This order of remanence acquisition is consistent with our observation that in all cases where A and B components are present, A is overprinted by the B component (Figure 6). This makes the reasonable assumption that both A and B remanences are thermoremanent magnetizations (TRM) which are best separated by thermal demagnetization. Constanzo-Alvarez and Dunlop [1988, Figure 14] report one example of a sample in which a B-overprinted-by-A bivectorial remanence is present, but we consider it to be suspect because the B component is retained only in very low coercivity domains (≤4 mT) and hence may be a spurious magnetization of VRM origin. Conversely their example of a sample in which the A is overprinted by a B remanence is more convincing although AF demagnetization is used, and is consistent with our observations. Also this eastward younging of the remanence is consistent with the eastern younging of isotopic age determinations derived from several different systems in the KSZ [Percival et al., 1988].

The pole position derived from the A component is at 33°E, 67°N (dp=6°, dm=9°, N=13). We agree with the interpretation of Constanzo-Alvarez and Dunlop [1988] that the A magnetization was acquired during uplift at 2.55 Ga during the Kenoran Orogeny, but consider their estimate of subsequent 15° to 30° WNW tilt to be too large. Recently, the Matachewan diabase (poles MD1-MD5, Figure 12) has been dated by the U/Pb zircon method at 2454±2 Ma [Heaman, 1988], which is much younger than the previously accepted age of 2.61 Ga [Gates and Hurley, 1973; Irving, 1979]. Thus the apparent polar wander path (APWP) used by Constanzo-Alvarez and Dunlop [1988] requires revision. Using our own estimate for the A direction, the oval of 95% confidence for the A pole (KGA) fits onto a revised APWP for tilt corrections about a N35° horizontal axis ranging from 0° to about 30° WNW, however, the best fit using minimum tilt correction is achieved at about 10°±5° (Figure 12). This value is in good agreement with the regional estimate of WNW downward tilt from geobarometry of 8°±2 [Percival and Card, 1983, 1985] and with the reflection seismic estimates of 20° decreasing to horizontal westward [West et al., 1989] and of about 10° [Percival, 1989]. Ernst and Halls [1984] found that the N-S trending Matachewan dikes at their sites 17 to 20, located between our sites 1 and 2, gave dips and remanence directions that suggest a 15° WNW tilt. More recent paleomagnetic work on the Matachewan dikes [Vandall and Symons, 1987; Halls and Palmer, 1990] has enlarged the data base, so that the Ernst and Halls [1984] dikes no longer require tilt correction to be concordant. Halls and Palmer [1990] now suggest WNW tilts of <10° for dikes

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that are more than 4 km west of the ILCZ. Thus we hold that the simplest interpretation is that the original A pole was located at about 59°E, 71°N (dp=6°, dm=9°, N=13) and a 10°±5° WNW tilt prior to 2.45 Ga moved it to its present position. The interpretation depends on the validity of the assumption of minimum apparent polar wander between the 2.55 and 2.45 Ga poles that was made to construct the APWP. We further suggest that the 30° WNW foliation dip observed in the KSZ is not a measure of the 10° WNW thrust fault dip, and thus that they reflect two discrete events.

Discussion of the B Component

The B component was identified at 18 accepted sites along the central and northeastern parts of the Highway 101 traverse. Constanzo-Alvarez and Dunlop [1988] found the B component at four sites along their Shawmere traverse in the Shawmere anorthosite. Only two of their site mean directions are represented by more than three specimens, and both fall within the directional limits of the B population found in this study (Table 2). Therefore all 20 accepted sites have been grouped together to give a mean direction of D=72.7°. I= -69.0° (k=47, α_{95}=4.8°, N=20).

The B component is found both in massive anorthosite and in a wide variety of granulite gneisses, and its direction is at an angle of about 40° to the foliation of the gneisses. Also the mean direction of D=73°, I= -69° (α_{95}=9°) for the nine B component sites from massive to slightly foliated rocks is indistinguishable from that of D=72°, I= -67° (α_{95}=7°) for the nine B component sites from moderately to well foliated rocks. Both factors suggest that the B remanence direction has not been biased by the fabric of the gneisses. Evidence for the B component being acquired after the A component was given above.

Although they have adjacent paleopoles (Figure 13b), the B component direction is statistically different from that of the 2.04 Ga Kapuskasing dikes [Halls and Palmer, 1990]. These dikes are nearly vertical, providing additional evidence that WNW tilt predates 2.04 Ga. The negative contact tests between the Kapuskasing dikes and the host granulite gneisses indicate that the B component was acquired during intrusion at 2.04 Ga or during an episode of regional metamorphism at some later date.

Based on its pole position, Constanzo-Alvarez and Dunlop [1988] interpret the B component to be a thermochemical overprint that was acquired at about 1.1 Ga during rapid uplift related to Keweenawan rifting and volcanism. We consider this unlikely. The 1.07 Ga Nemegosenda and 1.09 Ga Lackner Lake Complexes are the only substantial intrusions in the study area younger than 1.87 Ga, and they yield remanence directions of D=305°, I=46° (α_{95}=18°) and D=305°, I=64° (α_{95}=5°) [Symons and Garber, 1974; Symons, 1988]. Thus they are not antiparallel to the B direction. The 1.14 Ga [Stevens et al., 1982] lamprophyre dike at site 16 exhibits this same positive northwesterly direction. It also gives a positive contact test and an ambient temperature for the host rock of ≤200°C so that most of the regional uplift had substantially occurred by the 1.14 Ga date of intrusion. Further, preliminary paleomagnetic data from the 1.09 Ga Shenango

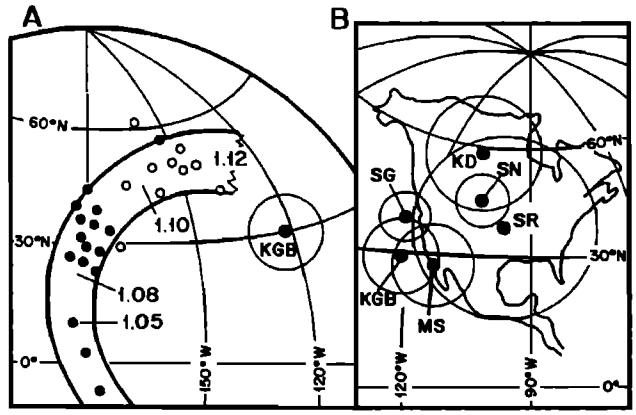


Fig. 13. Kapuskasing granulite B magnetization pole (KGB) at 124°W, 30°N (dp=7°, dm=8°) in relation to the APWP for (a) 1.12 to 1.00 Ga Keweenawan time [Halls and Pesonen, 1982; Palmer and Davis, 1987; Lewchuk and Symons, 1990] and (b) about 1.9 Ga for the Superior Province, where the poles are KD, Kapuskasing dikes, 2.04±0.01 Ga ($^{40}Ar/^{39}$ [Hanes et al., 1988]) [Halls and Palmer 1990]; MS, Molson dikes, 1.88±0.01 Ga (U/Pb zircon [Machado et al., 1986]) [Ermanovics and Fahrig, 1975]; SR, Spanish River complex, 1.84±0.10 (Rb/Sr [Sage, 1983]) [Robertson and Watkinson, 1974]; and SN, SG, Sudbury norite and granophyre, 1.85±0.01 Ga (U/Pb zircon [Krogh et al., 1984]) [Morris, 1984].

Complex about 50 km north of the study area indicate an antiparallel ESE and intermediate up direction [Dunlop et al., 1988] and the 1.09 Ga Clay-Howells Complex even farther to the north in the KSZ gives a direction of D=294°, I=27° (α_{95}=7°) [Lewchuk and Symons, 1990]. Neither result is similar to the B direction. In addition, there is no evidence for a pervasive 1.1 Ga thermochemical event in the KSZ that could totally reset the remanence in these host rocks on a regional scale and yet leave unaffected the K-Ar hornblende, K-Ar biotite and Rb-Sr biotite isotopic systems to record rapid uplift of the KSZ at about 1.93 Ga [Percival et al., 1988]. Percival et al. [1988] also pointed out that the minimum age for significant tectonic activity on the ILCZ and associated KSZ faults must predate intrusion of the 1.87 Ga Borden and 1.89 Ga Cargill complexes that cut KSZ fault planes without being significantly disturbed. There is also a NE trending diabase dike that gives a $^{40}Ar/^{39}Ar$ age of 1.5 Ga and cuts a splay of the ILCZ [Queen et al., 1989]. Finally, the B component pole does not fall on the well-defined 1.12 to 1.00 Ga or Keweenawan portion of the APWP (Figure 13a). Although a 30° WNW tilt correction brings the B component pole onto the APWP at about 1.09 Ga, such tilting is considered unlikely because the Nemegosenda and Lackner Lake Complex paleopoles are the same age and they do not show evidence of having been tilted.

We propose that the B component of the granulite gneisses and anorthosites was acquired after emplacement of the Kapuskasing dikes at about 2.04 Ga but prior to emplacement of the Borden and Cargill Complexes at about 1.87 Ga. Dunsmore and Symons [1990] have compared paleopo-

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les from the Slave and Superior Provinces and the intervening Trans-Hudson Orogen for the 2.4 to 1.7 Ga time period. They show that the data are consistent with a progressive closure of the Slave and Superior cratons to form the Trans-Hudson Orogen by about 1.8 Ga, as several Proterozoic plate tectonic models suggest [e.g., Gibb et al., 1980; Lewry, 1981; Green et al., 1985; Hoffman, 1988]. Thus the Slave and Superior Provinces must have separate APWP between 2.05 Ga and 1.85 Ga. The limited paleomagnetic data for the Superior Province for this time interval provide an excellent fit for the B component pole without tectonic correction at about 1.88 Ga, coeval with emplacement of the Sudbury Irruptive and the Molson dike swarm (Figure 13b). Correction for regional tilt would move the B component pole westward from the 1.85 to 1.88 Ga paleopoles to degrade the fit, and this suggests that only simple uplift has occurred since the acquisition of the B component. The proposed 1.88 Ga acquisition and uplift age agrees with the Proterozoic plate model of Hoffman [1988] who relates KSZ uplift to closure of the Trans-Hudson Orogen at this time.

Discussion of the C Component

The origin of the C component at the three sites along the northeastern part of the Highway 101 traverse is not clear. Its direction is entirely different from the proposed C component of Constanzo-Alvarez and Dunlop [1988]. The latter was found in 10 specimens from a total of six sites, none of which meet our acceptance criteria for site mean directions. The C component direction reported here gives a pole position of 119°E, 1°S (dp=7°, dm=8°, N=3) or an antipole at 61°W, 1°N. Neither pole lies on any defined portion of the APWP for the Superior Province before 1.6 Ga or for the North American craton thereafter [Irving 1979; Irving and Irving, 1982; Dunlop and Stirling, 1985; Dunsmore and Symons 1990]. Therefore the C component is likely to be either a hybrid magnetization of two components or a component that has undergone local tilting. The direction of the C component does not lie on a great circle between the A and B magnetizations, and therefore, it is unlikely to be an unresolved resultant of them. However it does lie perfectly on a great circle between the B component direction and a Keweenawan normal polarity direction of about D=295°, I=30°. The latter direction corresponds to an age of about 1085 Ma on the APWP in the midst of the long normal superchron of Halls and Pesonen [1982]. This is a likely time for intrusion of the normal polarity lamprophyre dikes and the Nemegosenda carbonatite, the closest large Keweenawan complex to the C component sites. Thus the C component could be a localized hybrid of the B component and a Keweenawan normal chemical remanent magnetization (CRM), both residing in similar high coercivity and unblocking temperature spectra. Alternatively the C component could be a tilted B component because the correction of C for a 40° WNW tilt aligns it almost perfectly with B. This alternative requires minor localized block faulting along subsidiaries of the ILCZ after 1.88 Ga. Evidence for both minor later faulting and localized hydrothermal effects is seen in minor fault offsets found in the 1.1 Ga Lackner Lake and Nemegosenda carbonatite complexes [Parsons, 1961], and in the 1.8 to 1.7 Ga and 1.2 to 1.0 Ga K/Ar and $^{40}Ar/^{39}Ar$ age dates found near the ILCZ [Archibald et al., 1986; Lopez Martinez and York, 1986].

Conclusion

Granulite gneisses and anorthosites on the western side of the KSZ were magnetized (A component) during uplift in the Kenoran Orogeny at about 2.55 Ga. This terrain was then tilted about 10°±5° WNW in response to thrust faulting on the ILCZ before emplacement of the Matachewan dikes at 2.45 Ga. After emplacement of the Kapuskasing dikes at 2.04 Ga, the eastern side of the KSZ was again uplifted rapidly on the ILCZ. This led to the acquisition of a second remanence (B component) at about 1.88 Ga, approximately concurrent with emplacement of the Borden and Cargill carbonatite complexes, and perhaps as a distant reflection of closure of the Trans-Hudson Orogen. A third magnetic direction (C component) is found at three sites near the ILCZ. It could be either a hybrid combination of the B component with a partial CRM acquired at 1.1Ga or the result of localized tilting of the B component by about 40° down to the WNW.

Acknowledgments. The authors wish to thank Alan Timmins, who assisted in the sampling and specimen preparation. David Dunlop, John Geissman, and an anonymous reviewer provided helpful comments on various drafts of this paper. The Natural Sciences and Engineering Research Council of Canada funded this research through Operating and Lithoprobe grants. Lithoprobe publication 160.

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(Received November 4, 1988;
revised April 16, 1990;
accepted April 16, 1990.)

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